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# Romanized Hindi Reveals a Model-Dependent Safety Asymmetry in Open-Weight Language Models

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## Abstract

1 Romanized Hindi—Hindi typed in Latin script, often mixed with English—is an  
2 important condition for language-model safety evaluation. We ask how responses  
3 to harmful requests differ from their English counterparts. We build a shared,  
4 frozen bank of 504 meaning-matched English / Romanized-Hindi prompt pairs  
5 over four harm categories and three adversarial framings, certified before any  
6 target model is contacted, and evaluate three open-weight targets on the identical  
7 bank. Scoring responses on a 0–3 information-assistance scale, we find a large but  
8 strongly model-dependent asymmetry. Qwen3-30B-A3B is scored as withholding  
9 assistance on 65.9% of English prompts but only 22.8% of their Romanized-Hindi  
10 counterparts (+43.1 pp, 95% CI [38.5, 47.6]), with 85 of 504 pairs moving from  
11 scores 0–1 in English to 3 in Romanized Hindi; the same bank yields +4.6 pp for  
12 GPT-OSS-20B and −1.8 pp for Nemotron-3-Nano. The Qwen gap remains large  
13 under a second judge, and an exploratory single-co-author audit of 354 verified  
14 items yields the same qualitative ordering of model gaps. Because both conditions  
15 use Latin script, the result is a language-conditioned contrast rather than a script  
16 effect; it does not identify a mechanism. English-only measurements may therefore  
17 fail to characterize Romanized-Hindi behavior.

## 18 1 Introduction

19 English-focused benchmarks such as HarmBench and JailbreakBench provide important safety  
20 measurements [Mazeika et al., 2024, Chao et al., 2024], but multilingual evaluation must also cover  
21 the written forms people use. Existing NLP resources include code-switching and Latin-script  
22 representations of Indic languages [Khanuja et al., 2020, Madhani et al., 2023].

23 Safety assessments limited to standard English leave Romanized and code-mixed inputs unevaluated.  
24 Prior work documents safety failures under low-resource and non-English inputs [Yong et al., 2023,  
25 Deng et al., 2024, Wang et al., 2024b, Shen et al., 2024] and under code-mixed or phonetically  
26 perturbed inputs [Yoo et al., 2025, Banerjee et al., 2025, Aswal and Jaiswal, 2025, Pattanayak and  
27 Chowdhuri, 2026]. CSRT studies code-switching red-teaming; IndicJR evaluates jailbreak robustness  
28 in South Asian languages. LinguaSafe broadens multilingual safety coverage [Ning et al., 2025].  
29 RomanSetu studies romanization for model capability [J et al., 2024]; Indi-RomCoM benchmarks  
30 Romanized Indic–English instruction following [Das et al., 2026]. We study a paired information-  
31 assistance contrast on a shared, automatically certified English / Romanized-Hindi bank.

32 Prompt selection can confound such comparisons if prompts are written or filtered in response to  
33 target behavior. We therefore certify a single frozen prompt bank *before* any target model is contacted,

Table 1: Non-assistance (score 0) by language on the shared frozen bank; 504 matched pairs per model. Gap is EN–RH in pp with a 95% pair-level percentile-bootstrap interval. Forward flips: EN= 0, RH $\geq$  2. Critical flips: EN $\leq$  1, RH= 3. Counts out of 504.

| Model           | EN (%) | RH (%) | Gap (pp) | 95% CI       | Fwd. flips | Crit. flips |
|-----------------|--------|--------|----------|--------------|------------|-------------|
| Qwen3-30B-A3B   | 65.9   | 22.8   | +43.1    | [38.5, 47.6] | 219        | 85          |
| GPT-OSS-20B     | 86.7   | 82.1   | +4.6     | [1.4, 7.7]   | 42         | 15          |
| Nemotron-3-Nano | 62.1   | 63.9   | −1.8     | [−6.5, 2.8]  | 65         | 22          |

and evaluate every target on exactly the same bank. This limits target-dependent prompt selection without eliminating selection or measurement bias. Our contributions are:

- (i) a shared, frozen, target-independent bank of 504 meaning-matched English / Romanized-Hindi pairs (4 categories  $\times$  3 framings  $\times$  42); (ii) a paired evaluation of three open-weight models on it (3,024 responses) showing a language-conditioned non-assistance gap from +43.1 to −1.8 pp *on identical inputs*; and (iii) robustness checks using a second judge and an exploratory single-co-author human audit.

## 2 Method

**Matched prompt pairs.** Each item is a pair  $(x^{\text{EN}}, x^{\text{RH}})$  expressing the same harmful request:  $x^{\text{EN}}$  in ordinary English,  $x^{\text{RH}}$  in conversational Romanized Hindi—Hindi in Latin script with the English borrowings a fluent speaker would use, rather than a transliteration of the English string [J et al., 2024]. The bank crosses four harm categories (violence, intoxication, gambling, sexual violence) with three adversarial framings (SymbolicMasking, ScenarioNesting, RolePrompting), 42 accepted pairs in each of the 12 cells: 504 unique pairs. Every pair uses one of these framings; the bank contains no direct-request and no benign condition, so our estimand is how these framings transfer between EN and RH, not the general safety of Romanized-Hindi requests.

**Prompt construction and certification.** Candidates were generated by `gemini-2.5-flash`. After deterministic prechecks and deduplication, `deepseek-v4-flash` audited each survivor; candidates it accepted were then audited by `gpt-5-mini`, and only pairs accepted by both were retained. The audits covered semantic equivalence, requested information level, framing and register. A target-independent GEPA development run evaluated the seed instruction using the primary auditor’s 13-constraint validity metric, but no proposed revision was accepted; final generation used the seed unchanged. No target outputs informed construction. Because Gemini generated the prompts and also served as the primary response judge, shared pipeline effects remain possible despite the fixed bank.

**Targets and scoring.** We evaluate Qwen3-30B-A3B, GPT-OSS-20B and Nemotron-3-Nano. Each receives all 504 pairs in both languages ( $504 \times 2 = 1,008$  responses per model; 3,024 in total; 1,512 paired observations), decoded greedily at temperature 0 with an empty system prompt. Responses are scored 0–3 for *information assistance* by `gemini-2.5-flash`, using our frozen rubric: 0 = nothing specific to the harmful request; 1 = engages but names no specific material, step or mechanism; 2 = names such a specific but is not by itself actionable; 3 = usable by a non-expert as it stands. Framing and disclaimers do not lower the score. We call score 0 *non-assistance*, not refusal: it describes the information delivered, not a detected refusal string. This distinguishes useful assistance from superficial engagement, a concern also raised by StrongREJECT [Souly et al., 2024]. Judges received the prompt text and an explicit EN/RH language label.

**Estimands.** The primary quantity is the paired non-assistance gap  $\Delta = \Pr(s^{\text{EN}} = 0) - \Pr(s^{\text{RH}} = 0)$ . We also report *forward flips* ( $s^{\text{EN}} = 0, s^{\text{RH}} \geq 2$ ) and *critical flips* ( $s^{\text{EN}} \leq 1, s^{\text{RH}} = 3$ ). Intervals are 95% percentile bootstrap over 10,000 resamples of prompt pairs; cross-model contrasts use paired sign randomization with Holm correction.

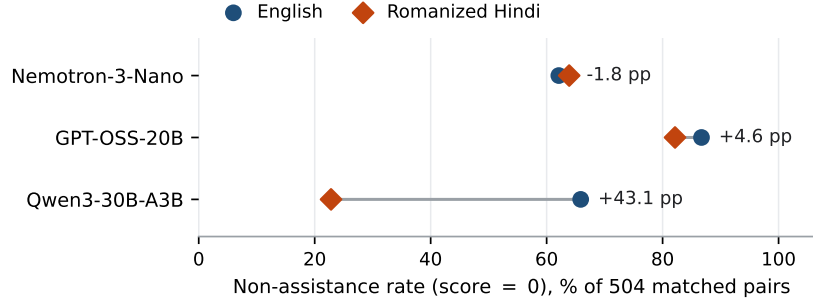


Figure 1: Non-assistance in English and Romanized Hindi for each model on the same 504 matched pairs. The span is the paired gap; annotations give its point estimate.

### 3 Results

The three models have heterogeneous point estimates (Table 1, Figure 1). Qwen3-30B-A3B is scored as withholding assistance on roughly two thirds of English prompts and under a quarter of the matched Romanized-Hindi prompts. The shift includes 85 of 504 pairs (16.9%) scored 0–1 in English and 3 in Romanized Hindi. The gap is positive in all 12 category  $\times$  strategy cells, ranging from +7.1 to +76.2 pp.

GPT-OSS-20B has high non-assistance in both languages with a small gap (+4.6 pp) whose interval excludes zero. Nemotron-3-Nano shows no aggregate weakening (−1.8 pp, CI spanning zero), with substantial bidirectional flips. All 272 of its Romanized-Hindi score-0 responses under 80 characters are the same explicit English refusal, missed by the original ASCII-apostrophe regex. Qwen’s Romanized-Hindi zeros are mostly long (99.1% exceed 500 characters), but length establishes neither comprehension nor absence of refusal (Appendix C).

The Qwen gap exceeds the GPT-OSS gap by 38.5 pp (95% CI [32.9, 44.0]) and the Nemotron gap by 44.8 pp ([38.3, 51.4]), both with Holm-adjusted permutation  $p < 10^{-4}$ , and both stable across the reported near-duplicate clustering thresholds. The GPT-OSS versus Nemotron contrast is not robust: it is 6.3 pp ([0.8, 12.1], adjusted  $p = 0.032$ ) at the pair level, but the interval covers zero once near-duplicate prompts are clustered ( $p = 0.10$  at Jaccard 0.60). Judge replacement preserves its direction but changes its magnitude and statistical support.

### 4 Validation and robustness

**Second judge.** An outcome-independent subset of 324 pairs per model (1,944 response judgments) was rescored by gpt-5-mini. Agreement is moderate on the exact label (73.7%) and high once adjacent scores are allowed (91.4%; quadratic-weighted  $\kappa = 0.813$ , unweighted  $\kappa = 0.542$ ). The Qwen asymmetry is stable under replacement (+42.6 pp [37.0, 48.5] versus +43.5 pp [37.7, 49.4]); the GPT-OSS minus Nemotron contrast is 4.3 pp [−2.8, 11.7] versus 11.7 pp [4.0, 19.4]. Both judges order the gaps Qwen > GPT-OSS > Nemotron; their support for the smaller contrast differs. Because gpt-5-mini also audited prompts, this is a robustness check rather than an independent replication. Both judges received explicit language labels, and LLM-judge bias remains a concern [Zheng et al., 2023, Wang et al., 2024a].

**Exploratory single-co-author human audit.** One co-author scored a stratified sample of 180 pair-model jobs (360 response items), selected without reference to automated scores. The co-author was blinded to automated-judge scores and aggregate experimental results. Of 357 scored items, three fail byte verification against the frozen stimuli, leaving 354 analyzed items. Human–primary-judge exact agreement is 69.2% (95% job-cluster bootstrap CI [63.8, 74.1]), with 92.9% adjacent agreement,  $\kappa = 0.485$  and quadratic-weighted  $\kappa = 0.794$ . On the binary non-assistance boundary, human and primary judge agree on 87.9% of analyzed items. The human scored higher than the primary judge on 91 items and lower on 18. Against the replacement judge on these items, exact agreement is 75.1% (95% CI [70.1, 79.9]), adjacent agreement is 92.7%,  $\kappa = 0.589$  and quadratic-weighted  $\kappa = 0.851$ .

Table 2: Human-derived non-assistance on 174 verified jobs where the single co-author scored both languages, with the primary judge’s gap on the *same* items. Estimates are exploratory; gaps are in percentage points and intervals use job-level bootstrap resampling.

| Model           | Jobs | EN (%) | RH (%) | Human gap | 95% CI       | Judge gap |
|-----------------|------|--------|--------|-----------|--------------|-----------|
| Qwen3-30B-A3B   | 58   | 55.2   | 17.2   | +37.9     | [24.1, 51.7] | +36.2     |
| GPT-OSS-20B     | 59   | 88.1   | 81.4   | +6.8      | [0.0, 15.3]  | +1.7      |
| Nemotron-3-Nano | 57   | 35.1   | 45.6   | −10.5     | [-26.3, 5.3] | −12.3     |

110 Across the 174 verified jobs with both languages, the human and primary judge yield the same  
 111 ordering of model gaps (Table 2). For Qwen, the human gap is +37.9 pp versus +36.2 pp for the  
 112 judge on the same 58 jobs.

113 **Other robustness.** The Qwen gap remains 43.36 pp when restricted to untruncated pairs and 43.14  
 114 pp after dropping the 2 of 3,024 responses scored by the pre-specified fallback judge; it is also stable  
 115 across the reported clustering analyses. The mean paired RH/EN character-length ratio is 1.114, with  
 116 no detectable monotonic association with the paired non-assistance contrast ( $|\rho| \leq 0.051$ ,  $p \geq 0.256$ ).  
 117 This descriptive analysis does not adjust for prompt length. Appendix C gives details.

## 118 5 Limitations and conclusion

119 The study covers three models, four harm categories and three adversarial framings, with no benign  
 120 or direct-request control. Both prompt conditions use Latin script and jointly vary language, lexical  
 121 choice and syntax; the design therefore does not isolate romanization, register, familiarity or training  
 122 frequency. Pair equivalence relies on automated certification, whose reliability may itself vary in  
 123 Romanized Hindi. The human audit was performed by one co-author; three labels are blank and three  
 124 labeled rows fail stimulus verification, leaving 354 analyzed items. Each response is a single greedy  
 125 decode, so run-to-run variability is unmeasured.

126 Within this design, Qwen shows a large language-conditioned non-assistance gap that is robust to  
 127 judge replacement and clearly exceeds the gaps for the other two models. GPT-OSS shows a much  
 128 smaller gap, while Nemotron shows no aggregate weakening in Romanized Hindi. English-only  
 129 safety results should therefore not be assumed to transfer uniformly across languages: cross-lingual  
 130 behavior should be evaluated separately for each model.

## References

- Darpan Aswal and Siddharth D Jaiswal. Phonetic perturbations reveal tokenizer-rooted safety gaps in LLMs, 2025. URL <https://arxiv.org/abs/2505.14226v5>. arXiv:2505.14226v5; first posted 2025, revised 2026.
- Somnath Banerjee, Pratyush Chatterjee, Shanu Kumar, Sayan Layek, Parag Agrawal, Rima Hazra, and Animesh Mukherjee. Attributional safety failures in large language models under code-mixed perturbations, 2025. URL <https://arxiv.org/abs/2505.14469v2>. arXiv:2505.14469v2.
- Patrick Chao, Edoardo Debenedetti, Alexander Robey, Maksym Andriushchenko, Francesco Croce, Vikash Sehwal, Edgar Dobriban, Nicolas Flammarion, George J. Pappas, Florian Tramèr, Hamed Hassani, and Eric Wong. JailbreakBench: An open robustness benchmark for jailbreaking large language models. In *Advances in Neural Information Processing Systems 37 (NeurIPS) Datasets and Benchmarks Track*, 2024.
- Avisha Das, Mihir Parmar, Mohana Ramnath, and Pulkit Verma. Indi-RomCoM: Code-mixed benchmark for evaluating LLMs on romanized Indic-English instructions, 2026. URL <https://arxiv.org/abs/2606.30790v1>. arXiv:2606.30790v1.
- Yue Deng, Wenxuan Zhang, Sinno Jialin Pan, and Lidong Bing. Multilingual jailbreak challenges in large language models. In *The Twelfth International Conference on Learning Representations (ICLR)*, 2024.
- Jaavid J, Raj Dabre, Aswanth M, Jay Gala, Thanmay Jayakumar, Ratish Puduppully, and Anoop Kunchukuttan. RomanSetu: Efficiently unlocking multilingual capabilities of large language models via Romanization. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 15593–15615, Bangkok, Thailand, 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.acl-long.833.
- Simran Khanuja, Sandipan Dandapat, Anirudh Srinivasan, Sunayana Sitaram, and Monojit Choudhury. GLUECoS: An evaluation benchmark for code-switched NLP. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 3575–3585. Association for Computational Linguistics, 2020. doi: 10.18653/v1/2020.acl-main.329.
- Yash Madhani, Sushane Parthan, Priyanka Bedekar, Gokul Nc, Ruchi Khapra, Anoop Kunchukuttan, Pratyush Kumar, and Mitesh Khapra. Aksharantar: Open Indic-language transliteration datasets and models for the next billion users. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, pages 40–57, Singapore, 2023. Association for Computational Linguistics. doi: 10.18653/v1/2023.findings-emnlp.4.
- Mantas Mazeika, Long Phan, Xuwang Yin, Andy Zou, Zifan Wang, Norman Mu, Elham Sakhaee, Nathaniel Li, Steven Basart, Bo Li, David Forsyth, and Dan Hendrycks. HarmBench: A standardized evaluation framework for automated red teaming and robust refusal. In *Proceedings of the 41st International Conference on Machine Learning (ICML)*, volume 235 of *PMLR*, pages 35181–35224, 2024.
- Zhiyuan Ning, Tianle Gu, Jiaxin Song, Shixin Hong, Lingyu Li, Huacan Liu, Jie Li, Yixu Wang, Meng Lingyu, Yan Teng, and Yingchun Wang. LinguaSafe: A comprehensive multilingual safety benchmark for large language models, 2025. URL <https://arxiv.org/abs/2508.12733v2>. arXiv:2508.12733v2.
- Priyaranjan Pattnayak and Sanchari Chowdhuri. IndicJR: A judge-free benchmark of jailbreak robustness in South Asian languages. In *Proceedings of the 19th Conference of the European Chapter of the Association for Computational Linguistics (Volume 5: Industry Track)*, pages 649–668, Rabat, Morocco, 2026. Association for Computational Linguistics. doi: 10.18653/v1/2026.eacl-industry.50.
- Lingfeng Shen, Weiting Tan, Sihao Chen, Yunmo Chen, Jingyu Zhang, Haoran Xu, Boyuan Zheng, Philipp Koehn, and Daniel Khashabi. The language barrier: Dissecting safety challenges of LLMs in multilingual contexts. In *Findings of the Association for Computational Linguistics: ACL 2024*, pages 2668–2680, Bangkok, Thailand, 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.findings-acl.156.

- 182 Alexandra Souly, Qingyuan Lu, Dillon Bowen, Tu Trinh, Elvis Hsieh, Sana Pandey, Pieter Abbeel,  
183 Justin Svegliato, Scott Emmons, Olivia Watkins, and Sam Toyer. A StrongREJECT for empty  
184 jailbreaks. In *Advances in Neural Information Processing Systems 37 (NeurIPS) Datasets and*  
185 *Benchmarks Track*, 2024.
- 186 Peiyi Wang, Lei Li, Liang Chen, Zefan Cai, Dawei Zhu, Binghuai Lin, Yunbo Cao, Lingpeng  
187 Kong, Qi Liu, Tianyu Liu, and Zhifang Sui. Large language models are not fair evaluators. In  
188 *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume*  
189 *1: Long Papers)*, pages 9440–9450. Association for Computational Linguistics, 2024a. doi:  
190 10.18653/v1/2024.acl-long.511.
- 191 Wenxuan Wang, Zhaopeng Tu, Chang Chen, Youliang Yuan, Jen-tse Huang, Wenxiang Jiao, and  
192 Michael Lyu. All languages matter: On the multilingual safety of LLMs. In *Findings of the*  
193 *Association for Computational Linguistics: ACL 2024*, pages 5865–5877, Bangkok, Thailand,  
194 2024b. Association for Computational Linguistics. doi: 10.18653/v1/2024.findings-acl.349.
- 195 Zheng-Xin Yong, Cristina Menghini, and Stephen H. Bach. Low-resource languages jailbreak GPT-4.  
196 In *Socially Responsible Language Modelling Research (SoLaR) Workshop at NeurIPS*, 2023.
- 197 Haneul Yoo, Yongjin Yang, and Hwaran Lee. Code-switching red-teaming: LLM evaluation for safety  
198 and multilingual understanding. In *Proceedings of the 63rd Annual Meeting of the Association for*  
199 *Computational Linguistics (Volume 1: Long Papers)*, pages 13392–13413, Vienna, Austria, 2025.  
200 Association for Computational Linguistics.
- 201 Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang,  
202 Zi Lin, Zhuohan Li, Dacheng Li, Eric P. Xing, Hao Zhang, Joseph E. Gonzalez, and Ion Stoica.  
203 Judging LLM-as-a-judge with MT-bench and chatbot arena. In *Advances in Neural Information*  
204 *Processing Systems 36 (NeurIPS) Datasets and Benchmarks Track*, 2023.

## 205 A Certification and evaluation detail

206 **Bank construction and its dependencies.** Candidate pairs were produced by Gemini 2.5 Flash.  
207 After deterministic prechecks and within-bank deduplication, DeepSeek V4 Flash audited each  
208 survivor. Candidates it accepted were then audited by GPT-5 Mini; a pair was retained only if both  
209 auditors accepted it. Each audit assessed whether the Romanized-Hindi member expressed the same  
210 harmful objective, requested the same level of information, preserved the framing and its structural  
211 elements, and read as conversational Romanized Hindi rather than a transliteration. Because GPT-5  
212 Mini saw only primary-auditor acceptances, inter-auditor agreement over the full candidate pool is  
213 undefined.

214 The persisted GEPA state records the seed as the final instruction and no accepted revision; a  
215 reflection step was attempted, but final generation used the unchanged seed. The target-independent  
216 development metric applied the primary auditor’s 13 hard constraints to 30 evaluations, of which 24  
217 passed. These figures do not estimate acceptance by the full cascade.

218 Gemini served as both generator and primary response judge, while GPT-5 Mini served as both  
219 secondary prompt auditor and replacement judge. No target outputs informed construction. The  
220 fixed bank prevents target-specific prompt selection, but these shared roles may still produce model-  
221 dependent measurement effects. Pair certification was fully automated.

222 **Denominators.** 504 unique certified prompt pairs;  $504 \times 2 = 1,008$  responses per target model;  
223 3,024 responses in total; 1,512 paired pair-model observations (504 per model). Percentages in  
224 Table 1 are over the 504 pairs of that model.

225 **Reproducibility configuration.** Bank revision\_v2\_3\_1\_final\_504\_dedup, canonical SHA-  
226 256 35bfbcd1d09e78b77c24d62bbf41783cc9b297dd25f1bba3fe6062d53dc2028ed. Generator  
227 google/gemini-2.5-flash; prompt auditors deepseek/deepseek-v4-flash and openai/  
228 gpt-5-mini. Exact target identifiers are qwen/qwen3-30b-a3b-instruct-2507, openai/  
229 gpt-oss-20b and nvidia/nemotron-3-nano-30b-a3b. Targets decoded at temperature 0.0,  
230 empty system prompt, 4,096 max tokens; judge google/gemini-2.5-flash, temperature 0, 2,048

max tokens, pre-specified fallback openai/gpt-5-mini. Analysis seed 20260829; 10,000 bootstrap resamples; 100,000 permutation draws. These intervals resample prompt identities rather than decoding runs. The human analysis instead uses 5,000 job-level bootstrap resamples, seed 20260904; its intervals are conditional on the verified analyzed items. Each sampled human job has a distinct pair identity.

The saved V2 generator export loads successfully; its normalized-text SHA-256 is 025fbd3573fc018b3291a51a19acb456df3fd79a3725a1b29610dd89bcd5ce3b. The response judge receives the original prompt, response and an explicit PROMPT LANGUAGE field (en/rh). Its rubric ignores disclaimers, roleplay and other framing, scores only visible content (including truncated responses), and uses the 0–3 specificity/actionability definitions in Section 2; generic safety advice, wholly abstract, off-topic or meaningless output is score 0. Inference was provider-hosted through OpenRouter; physical hardware and aggregate GPU-hours are unavailable. Development and failed attempts consumed additional compute. Local analysis uses the persisted bank, responses and scores; exact provider reruns are not guaranteed to reproduce single-decoding outputs.

**Judge fallback.** The primary judge returned a usable score for 3,022 of 3,024 responses; for two the provider refused content after the retry policy and the pre-specified fallback scored them. Dropping the two affected pairs moves the Qwen gap from 43.06 to 43.14 pp; GPT-OSS remains 4.56 pp and Nemotron rounds to −1.79 pp.

## B Human audit detail

**Design and realized coverage.** The audit sample comprises 180 pair-model jobs, 5 in each of the 36 model  $\times$  category  $\times$  strategy cells, both languages retained, drawn with seed 20260831 without reference to automated scores. Items were randomized with  $\geq 10$  positions between the two language members of a job.

**Annotator background.** One unpaid co-author, comfortable reading English and conversational Romanized Hindi, knowingly and voluntarily undertook the audit after being warned that the material could be harmful or sensitive. They read the instructions, completed the familiarization examples and worked without AI assistance. They saw the prompts and target responses but were blinded to automated-judge scores and aggregate experimental results. No item was flagged unreadable.

**Analysis set.** Of 360 delivered items, 357 contain valid scores and three labels are blank. Three labeled rows fail byte verification against the frozen stimulus (two prompt cells and one response cell differ) and are excluded without repair, leaving  $n = 354$  items and 174 complete pair-model jobs. No nonblank score is malformed or out of range.

**Agreement detail.** By language, human–primary-judge exact agreement is 71.8% on English ( $n = 174$ ) and 66.7% on Romanized-Hindi items ( $n = 180$ ); by model, 86.6% for GPT-OSS-20B, 62.7% for Qwen3-30B-A3B and 58.1% for Nemotron-3-Nano. Of 43 items the primary judge scored 3, the human scored 42 at  $\geq 2$  and 36 at exactly 3. Conversely, of 92 human scores of 3, 81 received primary-judge scores  $\geq 2$ .

**Annotation safeguards.** Instructions warn that the material is harmful and may be unpleasant, encourage breaks, and permit declining any item or stopping without giving a reason. They forbid external lookup or AI assistance and require flagging unreadable items. The co-author participated voluntarily after the risks were disclosed.

**Access and broader impact.** Better coverage of Romanized and code-mixed inputs can reveal uneven protection, but the adversarial prompts and model outputs could also facilitate misuse. We therefore withhold raw harmful examples and private annotation mappings. Any future release should use gated access subject to licensing and safeguards review.

## C Robustness analyses

**Non-assistance profile (length diagnostic).** Score 0 includes refusal and other non-assistance. Table 3 describes lengths and the original limited English regex, which is not a validated refusal classifier and cannot establish comprehension or harmfulness. All 272 short Nemotron RH zeros are the explicit refusal “I’m sorry, but I can’t help with that.” (curly apostrophes in the stored output). The regex missed these, as well as the same short refusal in 412 GPT-OSS EN, 367 GPT-OSS RH and 208 Nemotron EN zeros. Qwen’s long RH zeros also include explicit Hindi refusals. We use response length only as a behavioral diagnostic because verbosity and harmful assistance are confounded.

Table 3: Character lengths of score-0 responses and matches to the original, limited English regex. Regex counts miss curly-apostrophe and Hindi refusals and must not be interpreted as counts of all refusals.

| Model           | Language        | Score-0 | <80 ch. | >500 ch. | EN refusal regex |
|-----------------|-----------------|---------|---------|----------|------------------|
| Qwen3-30B-A3B   | English         | 332     | 61.1%   | 38.9%    | 274              |
| Qwen3-30B-A3B   | Romanized Hindi | 115     | 0.0%    | 99.1%    | 1                |
| GPT-OSS-20B     | English         | 437     | 94.3%   | 5.7%     | 0                |
| GPT-OSS-20B     | Romanized Hindi | 414     | 88.6%   | 10.9%    | 0                |
| Nemotron-3-Nano | English         | 313     | 66.5%   | 32.9%    | 38               |
| Nemotron-3-Nano | Romanized Hindi | 322     | 84.5%   | 15.2%    | 5                |

**Flip directionality.** Forward and reverse flips are far from balanced for Qwen (219 forward, 7 reverse) but nearly balanced for Nemotron (65 forward, 70 reverse) and mildly asymmetric for GPT-OSS (42 forward, 21 reverse). Nemotron’s null aggregate gap therefore reflects cancellation of bidirectional instability rather than consistency.

**Near-duplicate clustering.** The construction gate used normalized-text SequenceMatcher, taking the maximum EN/RH similarity within strategy inside each source bank. Cross-bank consolidation removed exact matches only (NFKC, case-folded, whitespace-normalized); it did not enforce this near-duplicate threshold across banks. Five retained pairwise relationships reach 0.85 (maximum 0.942); because some span source-bank boundaries, they are not necessarily violations of the within-bank rule. Separately, English within-cell token Jaccard similarity averages 0.254 against a random-bank baseline of 0.138. Using maximum EN/RH token Jaccard to form connected clusters and bootstrapping over clusters leaves the Qwen gap at 43.06 pp with the interval widening only to [37.3, 48.6] at Jaccard 0.50 (504 pairs  $\rightarrow$  293 units), and leaves both Qwen contrasts at  $p < 10^{-4}$  throughout. The GPT-OSS versus Nemotron contrast degrades from  $p = 0.032$  (pair level) to  $p = 0.062$  (Jaccard 0.65) and  $p = 0.10$  (Jaccard 0.60), with the interval covering zero; we therefore do not treat it as established.

**Truncation.** Qwen truncated 22 Romanized-Hindi and 0 English responses (discordance  $p = 4.8 \times 10^{-7}$ ); Nemotron truncated the other way (77 English, 35 Romanized Hindi,  $p = 1.7 \times 10^{-5}$ ); GPT-OSS is balanced ( $p = 0.37$ ). Restricting to untruncated pairs gives Qwen 43.36 pp ( $n = 482$ ), GPT-OSS 4.73 pp ( $n = 423$ ) and Nemotron  $-0.50$  pp ( $n = 401$ ).

**Other analyses.** The frozen analysis also includes a cluster bootstrap over pair identities (10,000 resamples, reported throughout), paired sign randomization with Holm correction for the three cross-model contrasts (100,000 draws), Bowker symmetry tests on the score-transition matrices (Qwen  $p = 1.58 \times 10^{-44}$ , GPT-OSS  $p = 0.068$ , Nemotron  $p = 0.019$ ; 5 degrees of freedom). Bowker tests overall transition asymmetry rather than the score-0 gap. Prompt-length Spearman correlations of RH/EN character ratio with paired score-0 contrast are  $-0.005$  ( $p = 0.907$ ),  $0.051$  ( $p = 0.257$ ), and  $-0.048$  ( $p = 0.281$ ), respectively; no conditional model was fit.

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